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CLINICAL REPORT

# Comparison of remimazolam and propofol in anesthetic management for awake craniotomy: a retrospective study

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## Abstract

A new ultra-short-acting benzodiazepine intravenous anesthetic agent, remimazolam, was launched in Japan in 2020. Anesthesia during awake craniotomy is reportedly being performed safely using remimazolam; however, studies on its efficacy in awake craniotomy have not been conducted. We aimed to compare the efficacy of remimazolam and propofol in awake craniotomy. In this retrospective study, patients who underwent awake craniotomy ($n=36$) at our hospital between December 2019 and January 2021 were divided into two groups: the propofol group (P group: $n=21$) and the remimazolam group (R group: $n=15$). There was no significant difference in the recovery time between the two groups ($p=0.18$). The number of patients experiencing nausea was higher in the R group than in the P group ($p=0.02$); however, regression analysis revealed that the use of remimazolam contributed to increased intraoperative nausea (odds ratio = 14.4, $p=0.04$). No significant differences were observed in the frequency of vomiting and other intraoperative complications between the two groups. In conclusion, remimazolam has the potential for use as an alternative drug in anesthetic management during awake craniotomy.

**Keywords** Remimazolam · Awake craniotomy · Propofol

## Introduction

Propofol has been widely used for intraoperative anesthesia management during awake craniotomy (AC) [1]. In Japan, AC is typically performed under general anesthesia, with patients waking during surgery for a language task (awake phase). When the tasks were completed, the patients are anesthetized again (Asleep–Awake–Asleep method: AAA). Remimazolam, a new ultra-short-acting benzodiazepine intravenous anesthetic agent [2], was launched in Japan in 2020. The anesthetic management for AC reportedly could be safely performed using remimazolam [3]. To date, no studies have compared the efficacy of remimazolam and propofol in anesthesia management. We, thus, compared the two drugs for intraoperative anesthesia management during AC.

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## Methods

This was a single-center, retrospective study approved by our ethics committee (Approval number: 2020-0601). The Ethics Committee waived the requirement for written informed consent because this study was retrospective research, so we have gained acquisition of participant informed consent by opt-out method. Patients could refuse and opt out of data being held in this study. Exclusion criteria were American Society of Anesthesiologists (ASA) classification above 3; having liver failure (Child–Pugh classification B or C); severe respiratory diseases (severe obstructive pulmonary disease); and refusal from the patient.

We investigated all AC cases ($n=37$) performed at our hospital between December 2019 and January 2021. One case was excluded because of using propofol with dexmedetomidine. Therefore, we divided the remaining cases ($n=36$) into two groups: the propofol group (P group: $n=21$) and the remimazolam group (R group: $n=15$).

General anesthesia was maintained via intravenous administration. In the patients in group R, general anesthesia was induced with remimazolam (Anerem 50 mg ® Mundipharma K. K. Tokyo, Japan) continuous infusion at 12 mg/kg/h and reduced to 1 mg/kg/h after the loss

of consciousness. In the P group, propofol (1% Diprivan Injection-kit®, aspenpharma Tokyo, Japan) was administered with 3 μg/mL of target control infusion (TCI). Fentanyl (50–100 μg) and remifentanil (0.1 μg/kg/min) were used for anesthesia induction in both groups. Anesthesia was supported with remimazolam (0.5–1.0 mg/kg/h) and propofol (2.4–3.5 μg/ml of TCI) in conjunction with remifentanil (0.1–0.15 μg/kg/min) to maintain the Bispectral monitor (BIS® Nihon-Kohden Tokyo, Japan) values between 40 and 60.

Supraglottic airway devices (SGA: I-gel®, Intersurgical, Berkshire, UK) were inserted in all the patients for airway management, and scalp blocks were also performed (supraorbital nerve and supratrochlear nerve, greater and lesser occipital nerves, auriculotemporal nerve, and zygomaticotemporal branch nerve).

Before the awake phase, we administered levetiracetam (500 mg) as an anti-epileptic agent with a low dose of dexamethasone (6.6 mg) as an antiemetic. All patients were given mannitol (150–300 ml). Once the dura was exposed, infusion of the anesthetic drug was terminated. After the patient was confirmed to be awake, recovery of spontaneous breathing (defined as respiratory rate > 10 breaths/min, tidal volume > 0.5 ml/kg) and the ability to respond to verbal commands (such as grip hands, dorsiflexion of the foot, and open eyes) as well as average BIS values of above 75 were observed, following which the SGA was removed. Patients in the R group were administered flumazenil upon recovery of spontaneous breathing, if they were not yet fully awake (unable to communicate or open their eyes). Metoclopramide (10 mg) was administered if the patients complained of nausea in the awake phase.

Further, if the patients complained of pain, a local anesthetic (1% lidocaine with epinephrine 1:100000) and flurbiprofen (50 mg), or a low dose of fentanyl (25 μg) were administered for analgesia. Patients were sedated with either remimazolam or propofol following tumor resection under AC.

The total amount of fentanyl administered before the awake phase and complications during the awake phase (vomiting, seizures, and agitation), and postoperative vomiting and nausea (PONV) were recorded. If the patients complained of headache during the awake phase, it was classified as intraoperative pain (Numeric Rating Scale: NRS > 4), and analgesic agents (local anesthetic, flurbiprofen, and fentanyl) were administered. The patients were classified as hypertensive (HT) if the blood pressure was > 20% of the baseline blood pressure or the systolic blood pressure was > 150 mmHg [4].PONV was defined as occurrences of nausea or vomiting until 24 h after the end of surgery. The primary endpoint was the time required for the awake intraoperative phase, defined as the time from discontinuation of anesthesia to SGA removal. The secondary endpoint was intraoperative complications, such as pain, seizure, nausea, vomiting, and delayed arousal.

All statistical analyses were performed using EZR [5]. All the data were analyzed using Fisher test and Mann–Whitney *U* test. Logistic regression analysis was performed to evaluate factors affecting intraoperative nausea. $p < 0.05$ was considered statistically significant.

## Results

There was no significant difference in the patient background characteristics between the two groups and no significant difference in the time required for arousal between the two groups ($p = 0.18$).

In group P, there was one patient who could not be aroused; thus, we abandoned AC 40 min following discontinuation of anesthesia (P group $n = 21$). In group R, 8 of the 15 patients who were treated with 0.2 mg flumazenil demonstrated good arousal (time required for spontaneous breathing to recover: $14.8 \pm 2.6$ min). No patient showed agitation or seizure and none of them were re-sedated after administration of flumazenil.

The frequency of nausea during awakening was significantly higher in the R group ($p = 0.02$), and the regression analysis revealed that the use of remimazolam contributed to increased intraoperative nausea (odds ratio = 14.4; 95% confidence interval: 1.23–186, $p = 0.04$).

There were no significant differences in the frequency of vomiting, pain, agitation, seizures, HT, and PONV between the two groups (Table 1).

## Discussion

For proper intraoperative management, the patient should be clearly awake during AC. The combination of propofol and remifentanil has ideal pharmacokinetics for AC [6]. However, no antagonist of propofol exists; thus, some patients could not undergo AC due to delayed arousal.

Remimazolam has been developed, which is a novel ultra-short-acting benzodiazepine agent with fast onset, short duration of sedative action, and rapid recovery, compared to the available benzodiazepines (e.g., midazolam) [1, 7].

Therefore, remimazolam has the potential for use in anesthetic management during AC, as an alternative to propofol. Flumazenil can antagonize the anesthetic effect of remimazolam; thus, if patients anesthetized with remimazolam experience delayed arousal, flumazenil can be injected to enable good arousal to ensure that the operation is performed. Based on a study, which reported re-sleeping of patients following reversal of remimazolam by flumazenil [8], we recommend flumazenil to be administered after

**Table 1** Patient data

| | Group P ($n=21$) | Group R ($n=15$) | $p$ value |
|---|---|---|---|
| Age (years) | 44.9 ± 14.6 | 44 ± 6.07 | 0.86 |
| Sex (male/female) | 9/12 | 11/4 | 0.1 |
| Height (cm) | 164.8 ± 9.77 | 163.2 ± 8.18 | 0.5 |
| Weight (kg) | 61.9 ± 12.8 | 67.5 ± 14.4 | 0.32 |
| The amount of fentanyl before awake phase (μg) | 150 ± 45.9 | 175 ± 44.5 | 0.49 |
| HT | 9 | 7 | 1.0 |
| Nausea | 1 | 6 | 0.02 |
| Vomiting | 1 | 3 | 0.37 |
| Pain | 8 | 6 | 1.0 |
| Seizure | 2 | 1 | 1.0 |
| Agitation | 2 | 1 | 1.0 |
| Awake time (min) | 109 ± 36.5 | 112 ± 38.9 | 0.65 |
| Time required for arousal (min) | 19 ± 7.33 | 21 ± 6.06 | 0.18 |
| PONV | 8 | 7 | 0.73 |

Values are presented as mean ± standard deviation or number of patients

Group P: Propofol group, Group R: Remimazolam group. A $p$ value less than 0.05 was considered statistically significant

*HT* hypertension, *PONV* postoperative nausea and vomiting

confirming the full recovery of the patients' spontaneous ventilation not to be de-saturating in AC.

In our study, nausea frequency was significantly higher in the R group. In the awake phase, patients often experience nausea and vomiting because of anesthetic agent effect, the psychological condition of the patient (intraoperative anxiety and fear), and inadequate analgesia of the surgical site, extension of dura mater and meningeal vessels [9, 10]. Propofol was formerly reported with evidence that it possesses direct antiemetic effects and prevents postoperative nausea and vomiting [11]. Remimazolam does not reduce perioperative nausea and might not have the antiemetic effects that propofol has. Therefore, remimazolam probably resulted in more nausea events than propofol in our research.

Although nausea was relieved in all patients by administering antiemetic agents (metoclopramide 10 mg), intraoperative nausea was of greater significance when remimazolam was used during AC. If required, fentanyl administration could be reduced before the awake phase or additional prophylactic antiemetic agents, such as 5-hydroxytryptamine (5HT) blockers, could be administered.

A major limitation of this study was its retrospective nature. Because of the use of single-institution retrospective data, the limited sample sizes and resulting low power increased the probability of type 2 error. A power analysis was performed using the calculated frequencies for intraoperative nausea in this investigation (R group: 40% and P group: 4.7%), which showed that 54 patients would be necessary to reveal a statistically significant difference with adequate power (80%) based on $\alpha=0.05$. Hence, further prospective studies are warranted in the future.

In conclusion, remimazolam is a novel alternative in anesthetic management during AC.

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## Declarations

**Conflict of interest** None.

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